Power to progress: Electricity Supply, Inequality and Development in Rural India

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Abstract

Does the same infrastructure improvement deliver the same development gains to communities held back by poverty as to those held back by social marginalization? I study this question for electricity reliability in rural Bihar, exploiting the structure of distribution networks in which supply is allocated at the feeder level so that all villages on a feeder receive the same hours. I use the caste and the income composition of the feeder as two instruments for village electricity supply. The first stage shows that feeders serving a higher share of Scheduled Caste/Scheduled Tribe or low-income villages receive fewer supply hours. The central finding is that reliability produces *unequal* returns across forms of disadvantage. Economically disadvantaged villages translate greater electricity supply into gains in both farm and non-farm activity, together with positive effects on wealth. Caste-disadvantaged villages, facing the same institutional supply mechanism, exhibit gains concentrated in agriculture, with weaker non-farm outcomes and no corresponding wealth gains. A simple model rationalizes the pattern: electricity relaxes a *material* constraint that is binding on the income-poor but cannot relax the *social* constraint, exclusion from credit, markets, and non-farm occupations, that binds on the caste-marginalized. Infrastructure reliability alone may therefore be insufficient to reduce inequality when access to complementary opportunities remains socially constrained.

Keywords: electricity, infrastructure, inequality, caste, public services, rural development

1 Introduction

Infrastructure is widely expected to be a great equalizer: extend a road, a grid, or a pipe to a disadvantaged community and it gains the same productive opportunities as everyone

else. Yet disadvantaged communities are not homogeneous, and the constraints they face are not interchangeable. A village can be poor because it lacks material resources, or it can be marginalized because social hierarchy restricts its access to credit, markets, and occupations. This paper asks whether a single infrastructure improvement in the form of greater electricity reliability, generates the same development gains across these two forms of disadvantage. The answer turns out to be no, and the divergence is informative about when infrastructure reduces inequality and when it does not.

The motivation for focusing on electricity is well known. About 750 million people worldwide still lack access (International Energy Agency (2023)), and electricity consumption is strongly correlated with GDP per capita (Burke et al. (2018)). India has electrified rapidly through programs such as the Rajiv Gandhi Grameen Vidyutikaran Yojana (RGGVY), Deendayal Upadhyaya Gram Jyoti Yojana (DDUGJY), and Saubhagya, raising the access rate from 70% to 93% in a decade (Routley (2019)). But the causal impact of electrification on development remains contested (Lee et al. (2020)). Studies report effects on manufacturing output (Peters et al. (2011), Kline and Moretti (2014)), female labor force participation (Dinkelman (2011), Grogan and Sadanand (2013)), child education (Khandker et al. (2014), Akpandjar and Kitchens (2017)), respiratory health (Barron and Torero (2017)), and income (Rao (2018)); others find limited or no effects on women's employment (Grogan (2016), Burlig and Preonas (2016)), income (Bos et al. (2018)), health and education (Lenz et al. (2017)), or non-agricultural employment (Lewis and Severnini (2020)). A recurring theme is that effects vary by income group (Niu et al. (2013), Lee et al. (2020)). Much less is known about how effects vary by *social* position, or whether the relevant margin is access at all rather than the reliability of supply: even after a village is "electrified," daily hours often remain low and erratic.

I study these questions among the poorest and most marginalized communities in India - Scheduled Caste (SC) communities in Bihar, one of the country's poorest states. Caste-based segregation has been a persistent feature of rural India (Srinivas (1987)), residential segregation by caste exceeds segregation by socio-economic status (Vithayathil and Singh (2012)), and in segregated areas minorities receive worse access to public goods (Banerjee and Somanathan (2007), Bali and Paul (2021), Asher et al. (2024)). SC communities remain among the most disadvantaged despite affirmative action (Hnatkovska et al. (2012), Asher et al. (2020)). This combination of deep disadvantage and well-documented social segregation is exactly what makes Bihar a useful laboratory for separating material from social constraints.

The empirical strategy exploits the physical structure of electricity distribution. Supply in rural Bihar is managed at the feeder level: all villages connected to the same feeder receive the

same hours of electricity, and which feeders are energized when supply is rationed is decided at the power substation. Because feeders aggregate several proximate villages of differing composition, the social and economic make-up of the *feeder*, not of the village's own choosing, shifts the supply a village receives. I use the caste composition and, in parallel, the income composition of the feeder as two instruments for village supply hours.

The paper's central result is a contrast. Under the income-based instrument, greater reliability produces broad-based gains: farm activity, non-farm activity, and village wealth all rise, consistent with electricity relaxing a binding material constraint. Under the caste-based instrument, the *same* institutional improvement raises agricultural activity but is accompanied by weaker non-farm outcomes and no comparable wealth gains. A simple two-sector model in Section 3 shows why a single mechanism can produce both patterns: electricity raises the return to a locally available technology (groundwater irrigation) for everyone, but converting that into non-farm expansion and wealth requires complementary access i.e. credit, markets, networks, entry into non-segmented occupations, that electricity does not provide. Income poverty is a constraint electricity can help relax; social marginalization is not.

The supporting results establish the building blocks of this contrast. Pooling across villages, more reliable supply increases agricultural activity measure by higher irrigated-land shares and larger areas sown in both the kharif and rabi seasons, and raises clean-energy adoption, with limited short-run evidence of structural shifts toward non-farm employment or of health gains, and only modest education effects (a reduction in primary-grade repetition). The mechanism is irrigation: Bihar's groundwater pumping depends on electricity, so reliability primarily moves agricultural margins.

This paper makes three contributions. First and most important, it provides causal evidence that the returns to infrastructure reliability differ across *forms* of disadvantage, not merely across income levels, social marginalization, distinct from poverty, limits the conversion of infrastructure into broad-based advancement. This sharpens a literature on heterogeneous infrastructure returns that has focused largely on average effects or income heterogeneity. Second, it documents that the *allocation* of reliability itself tracks social composition: feeders serving more SC/ST or lower-income villages receive fewer hours, evidence that infrastructure provision reflects underlying social inequalities. Third, it shows that where reliability does bind, it operates through agricultural production and household energy use, with slower responses in health and education.

The rest of the paper is structured as follows: Section 2 describes the institutional context and political economy of electricity supply in Bihar. Section 3 develops the conceptual framework and

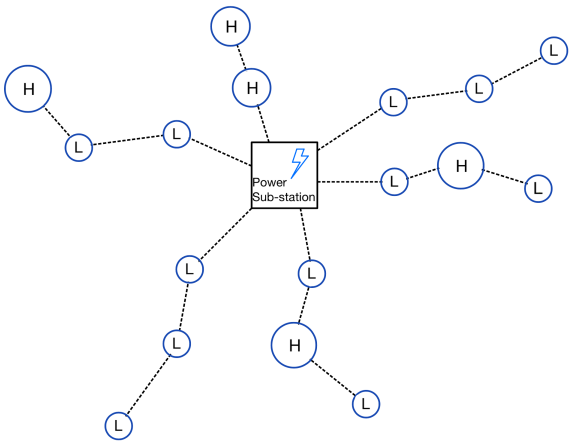


Figure 1: Feeder structure

the model of unequal returns. Data is described in Section 4. Section 5 presents the identification strategy; Section 6 reports the pooled (building-block) results; Section 7, the heart of the paper, compares caste- and income-based disadvantage. Section 8 concludes.

2 Background

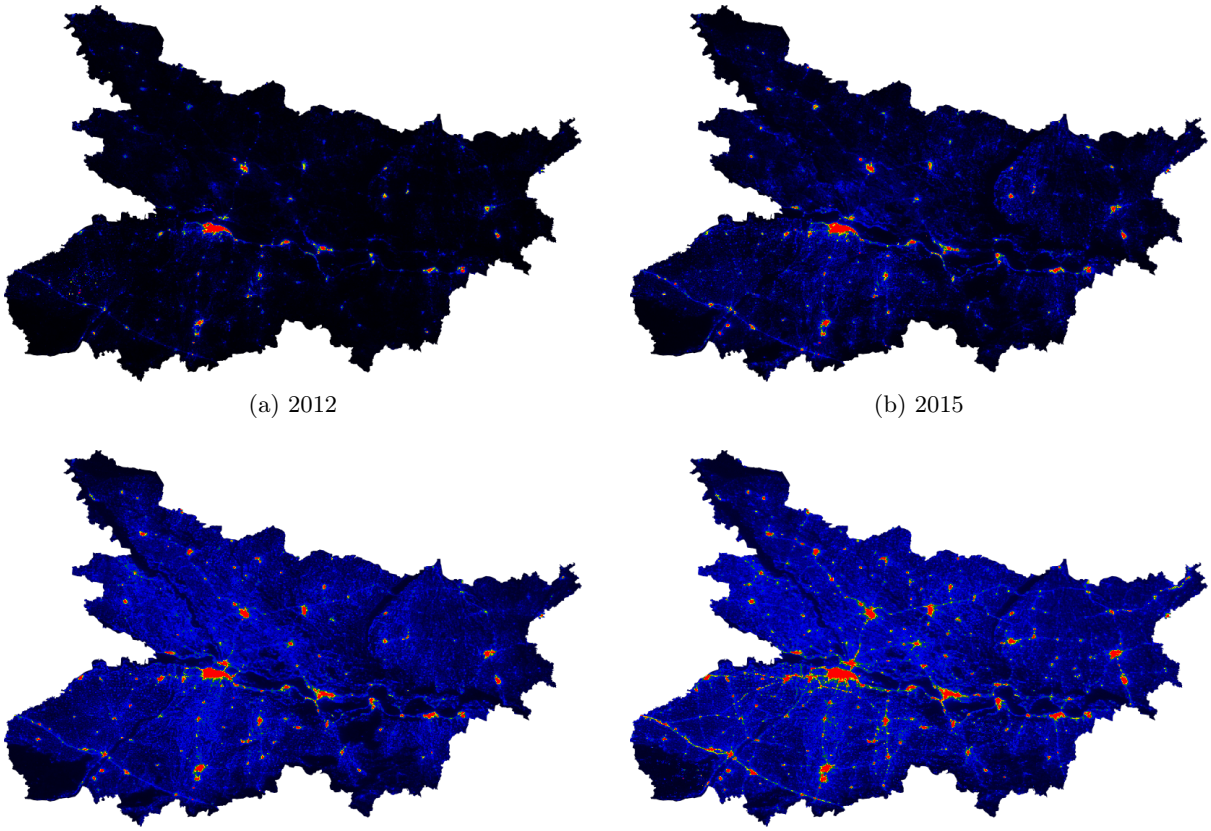
2.1 Electricity supply infrastructure

Electricity supply in Bihar operates through Power Substations (PSSs). Each PSS has several feeders (supply lines) that deliver electricity to one or more villages. These feeders normally go in a particular direction from the PSS and continue connecting villages in the same direction.

Supply can be controlled at the feeder level, meaning all villages connected to the same feeder receive the same number of hours of electricity. Figure 1 shows a simplified version of this setup. The Power substation supplies to the area around it using power lines called feeders. The switches for which feeders get turned on and which don't are located at the PSS. Once a feeder gets turned on, all the villages connected to it get electricity supply. These villages are different in terms of their caste-composition and income-composition, etc. The labels show low caste (income) villages (L) and high caste (income) villages (H). The caste composition of the feeder matters for the decision on how much supply all the villages on the feeder would get due to the inherent biases present in the decision-making process.

2.2 Progess on electricity in Bihar

Electricity access in Bihar increased from 17% in 1993 to over 96% by 2019-21 (Routley (2019)). Over half the villages in Bihar were not electrified as per the 2001 census. The govern-



(a) 2012 (b) 2015

(c) 2020 (d) 2025

Figure 2: Nightlights intensity

ment implemented several electrification programs, mostly funded by the central government, with the aim to electrify all the villages in the state. This project gained momentum in Bihar in around 2011-12, when two state electricity distribution utilities were established. 100% rural electrification was declared in October, 2018. Overall, the expansion in electrification in Bihar over the last decade and a half has been more of a supply side story. Urban areas now get 22-24 hours of supply per day while rural areas also get over 20 hours (BERC).

This makes Bihar an ideal place to study this. A lot of villages were connected in a relatively short span of time. And in the last 2-3 years, the focus has shifted to improving supply hours by generating more electricity. So, we get a period of increased electrification with limited supply. And another of gradual increase in supply. Figure 2 shows the progress in terms of night lights intensity.

2.3 Caste system

The caste system in India is deeply hierarchical, with Scheduled Castes (Dalits) at the bottom, historically excluded and oppressed (Ambedkar (1936), Kak (1993), Vani et al. (2014)). Figure 3 shows the caste hierarchy.

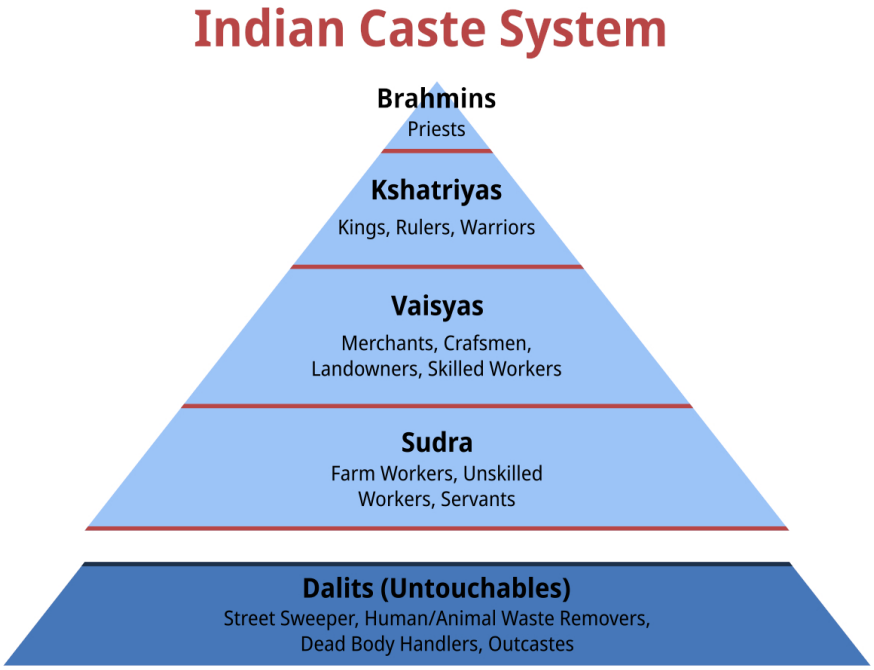


Figure 3: Caste hierarchy

Residential segregation by caste is more prominent than level of segregation by socio-economic status (Vithayathil and Singh (2012)). There is evidence of caste-based disparities in access to public services (Banerjee and Somanathan (2007), Asher et al. (2024)). SC communities are among the most socioeconomically disadvantaged, and although the disadvantages have been reduced due to affirmative action programs, differences still remain (Hnatkovska et al. (2012), Asher et al. (2020)). SCs and STs continue to be the most disadvantaged groups, with 42.93 percent and 42.7 percent of households living below the poverty line, respectively (Government of Bihar (2023)). The SC-ST population is fairly spread out in terms of geography. Figure 4 shows the villages in Bihar based on the proportion of SC-ST population within them.

2.4 Political economy of electricity supply

When overall supply of electricity is limited, a decision has to be made on who to prioritize. Khanna and Rowe (2024) show that in urban Delhi, feeders with a higher proportion of urban consumers get priority over feeders with a lower proportion. The reason behind that is that the industrial consumers pay a higher tariff, which also ends up cross-subsidizing the domestic consumers. The distribution companies want to increase their revenues and it makes financial sense to increase supply where they are expected to get highest price.

In rural areas, however, caste plays a significant role in access decisions. Asher et al.

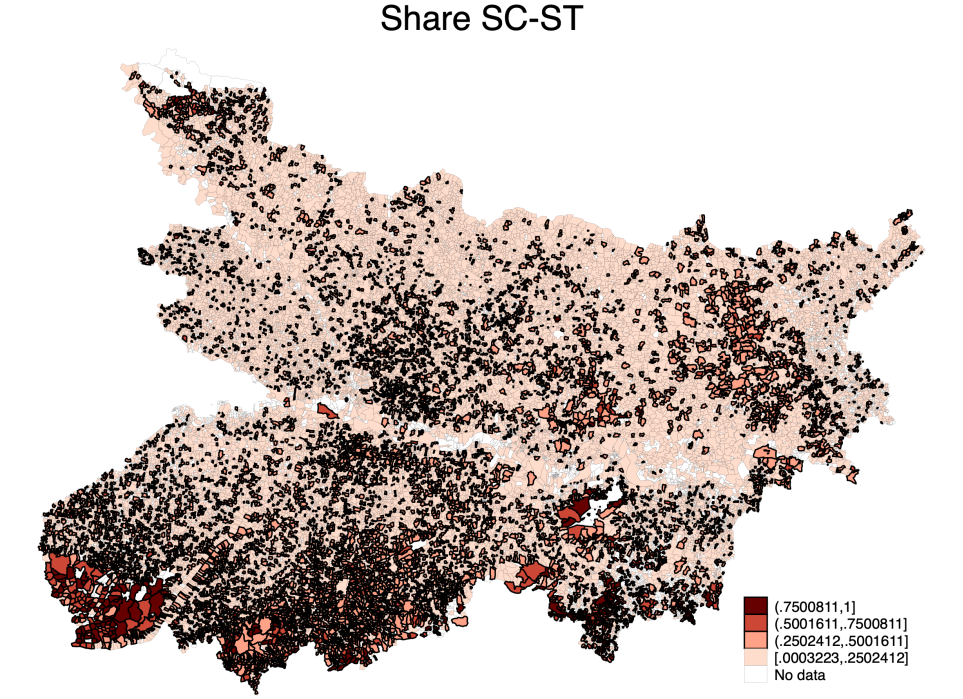


Figure 4: Share of SC-ST population in village
Source:SECC, 2011

(2024) highlight that for electricity 'SCs (who are somewhat poorer on average) face worse neighborhood-level disadvantages.' Aklin et al. (2021) also show that increasing the SC population by 10 percentage points reduces the probability of RGGVY implementation by 2 percentage points in Uttar Pradesh, the neighbouring state of Bihar. So, SC villages are worse off both on electrification as well as on supply after electrification.

Local politicians as well as bureaucrats might have a preference for upper caste villages for political as well as social reasons. This means that when supply is limited, they are the ones whose power gets cut. There is anecdotal evidence on repairs also taking longer for SC/ST majority villages. As caste is highly correlated with villages, even an income based supply allocation schedule would be expected to be worse for the lower caste villages.

3 Conceptual framework

3.1 Electricity and development outcomes

Electricity is a key input into rural development, but its effects depend on both the availability and the reliability of supply. Reliable electricity can raise human capital (lighting for study, electronic learning tools, better school and clinic infrastructure), improve health (refrigeration for vaccines and medicines, reduced reliance on polluting fuels), and, most relevant here, raise productivity by powering irrigation pumps and small enterprises, potentially shifting labor

from subsistence agriculture toward higher-productivity activities. Let village-level outcomes Y_i (education, health, employment, economic activity) depend on reliability S_i and village characteristics X_i, $Y_i = f(S_i, X_i)$. The empirical question is not only whether $\partial Y_i/\partial S_i > 0$ on average, but whether it is the *same* for villages disadvantaged along different dimensions. The next subsection makes that question precise.

3.2 A model of electricity, complementary inputs, and unequal returns

The effect of reliability depends on a village's ability to *use* electricity productively. Consider a representative village producer with one unit of productive effort allocated between a farm and a non-farm sector; let $n \in [0, 1]$ be the non-farm share and S electricity reliability.

Farm output uses electricity for groundwater irrigation, a *locally available* technology requiring only a pump and an aquifer:

$$q_F = \theta_F \, S^{\phi} \, f(1-n), \qquad \phi \in (0,1), \; f' > 0, \; f'' < 0.$$

Non-farm output also uses electricity but requires a fixed setup cost κ (equipment, working capital, a foothold in a market) that must be financed and sold into:

$$q_N = \mathbf{1}\{\text{entry}\} \cdot \big[A \, \theta_N \, S^{\psi} \, h(n) - \kappa \big], \qquad h' > 0, \; h'' < 0.$$

The important element here is A: *access to the complementary inputs that make non-farm activity viable* - credit to finance κ, markets to sell into, transport, networks, and freedom from occupational segmentation. The producer maximizes $q_F + q_N$. Wealth accumulates out of *monetized* surplus, which comes disproportionately from the non-farm sector (farm output is partly subsistence-consumed), so $\Delta W \propto \Delta q_N$.

The two forms of disadvantage enter through *different* parameters:

- **Income disadvantage**: low capital, so the household initially cannot pay κ; but A is normal. The door to non-farm activity is shut by a lack of cash.
- **Caste disadvantage**: low A (exclusion from credit, markets, networks). The door is shut by social exclusion, and crucially $\partial A/\partial S = 0$, soelectricity does nothing to relax caste-based exclusion.

Proposition 1 (Farm gains are universal)**.** *Holding n fixed, $\partial q_F/\partial S = \phi \, \theta_F S^{\phi-1} f(1-n) > 0$ for every village, regardless of A or capital. Irrigation is a local channel, so both disadvantaged groups raise farm output with S.*

This matches fig. 13, where farm coefficients are similar across the caste and income specifications.

Proposition 2 (The non-farm margin is gated by A, not by S)**.** *Non-farm entry is profitable only when $A\,\theta_N S^{\psi} h(n^*) > \kappa$, defining a viability threshold $A^*(S)$. If $A \geq A^*$, electricity raises the non-farm return and reallocates effort toward non-farm. If $A < A^*$, non-farm stays at the corner and electricity, by raising only the farm return, pulls effort* further into agriculture*, so $\partial q_N / \partial S \leq 0$. When farm is the more electricity-complementary use ($\phi > \psi$), $A^*(S) \propto S^{\phi-\psi}$ is* increasing *in S: more reliable power raises the bar for non-farm viability in low-A villages.*

The model thus *predicts* the negative non-farm coefficient for caste villages rather than leaving it a puzzle.

Proposition 3 (Wealth runs through the non-farm channel)**.** *Since $\Delta W \propto \Delta q_N$, electricity raises wealth only where the non-farm margin is open: $\partial W / \partial S > 0 \iff A \geq A^*$. Farm gains alone do not register in wealth because they are largely subsistence-consumed or reinvested in farming.*

Proposition 4 (The contrast)**.** *Combining the above: an* income-poor *village ($A \geq A^*$, low capital) sees farm output rise (P1), which finances κ, opening the non-farm margin (P2) and raising wealth (P3) - broad-based gains. A* caste-marginalized *village ($A < A^*$) sees farm output rise (P1), but the door stays locked because $\partial A / \partial S = 0$ (P2) and wealth is flat (P3) - agriculture-only gains.*

The model has three implications worth stating. It *unifies* the otherwise separate results of Sections 6 and 7 under one mechanism; it is *falsifiable* i.e. it is wrong if caste villages show non-farm or wealth gains, or if farm effects differ sharply across the two groups; and it recasts the headline as a structural statement: the divergence is not "caste is worse" but "electricity relaxes the material constraint κ (via income) and not the social constraint A."

3.3 Empirical implications

The framework motivates estimating

$$Y_{i,t} = \alpha + \beta S_{i,t} + X_i \delta + \epsilon_{i,t},$$

where β is the causal effect of reliability. Direct estimation is confounded because supply is correlated with local economic conditions, infrastructure, and political influence. I instrument

$S_{i,t}$ with the social (and, separately, the economic) composition of the feeder serving village i. Letting u_i denote the SC/ST share connected to the feeder,

$$S_{i,t} = \gamma_0 + \gamma_1 u_i + X_i\theta + \mu_{i,t}.$$

4 Data

4.1 Socio-Economic Caste Census (SECC) 2011

The socio-economic caste census of 2011 (Government of India (2016)) has information on total population, caste composition, baseline poverty, distance from nearest district/town, and availability of several public services. It also has data on agricultural and other employment, as well as education levels for each village in Bihar. These variables are used to construct baseline village characteristics and control variables in the empirical analysis.

4.2 Feeder-village mapping

The electricity department in Bihar did not have data on feeder-village mapping directly. But they shared their anonymised consumer list which has information on the feeder that the consumer is connected to and the village column from their address. This forms the base for mapping feeders to villages, as well as get a measure of the number of consumers in each village.

4.3 SHRUG

Using the SHRUG dataset developed by Asher et al. (2021), we obtain village-level measures of nightlights, wealth, and amenities. I am also getting data on relative wealth index in 2016. This is one of the outcomes that I have in my analysis. Other features include PM2.5 measures from 2010 to 2023, average nightlights from 2012 to 2020. It also has data on village level amenities in 2019 along with some data on health, employment and education in 2019. These have been used as outcomes.

4.4 UDISE+

The Unified District Information System for Education Plus (UDISE+) is the primary administrative database on school education in India, maintained by the Ministry of Education Department of School Education and Literacy (2023). It collects annual information from all recognized schools and includes data on enrollment, grade progression, teachers, and school infrastructure. In this study, UDISE+ is used to construct village-level measures of educational

outcomes, including proportion of students who repeat each class.

4.5 Predictions using satellite data

In the absence of admin data on village level supply hours, I used the methodology developed by Min and Gaba (2017) and Mann et al. (2016b), coupled with actual supply hours for some locations in Bihar between 2014 and 2017 from Prayas Energy Group (2015) and data on village characteristics from SECC, 2011, I use a random-forest and decision-tree based machine learning algorithm to predict supply hours at village level from 2012 to 2023. To increase the precision of the predictions, especially at low supply levels, I also add the villages which were not electrified till some point in the timeframe. The prediction models achieve high explanatory power, with R^2 values ranging from approximately 0.80 to 0.92. These predicted supply hours serve as a proxy for electricity reliability at the village level. Appendix C explains the prediction algorithm in more detail.

I also extracted information for irrigation practices at village levels from google earth engine (Gorelick et al. (2017)).

4.6 Descriptive statistics

Table 1 presents baseline characteristics for villages in Bihar. The boundaries of the villages is taken as per the 2011 census. It can be seen that about 63% of the villages had power supply available in 2011. This was up from just over half in 2001 (Census, 2001). Over the next 6-7 years, all the remaining villages were electrified. The average hours of supply was quite low though, about 3 hours a day in summers and less than 3.5 hours in the winter.

The villages have on average 465 households, and the poverty rate is quite high at 42%. The average SC-ST share is around 21%, which is in line with the current population estimates as well. The share of SC-ST population has not changed significantly over the last decade and a half. The average village is about 33 kms from the district headquarters (HQ) and about 10 km for their block HQ. Availability of telephones, post offices, and community health centers were low. About 64% of the villages were connected with a pucca road though, and about 70% with an all-weather road.

5 Identification strategy

Due to potential endogeneities, I am using the caste-composition of the feeder as an instrument for electricity supply. Using the Socio-Economic and Caste Census (SECC), 2011, I find

Table 1: Baseline Characteristics (2011)

	mean	sd	min	max	count
2013 irrigation index	0.168	0.023	-0.033	0.264	38611
DMSP nightlights mean	3.290	3.497	1.896	59.521	38609
viirs nightlights estimated	0.234	0.497	0.094	11.498	38609
pm25 mean	87.900	16.548	41.190	117.615	38609
Rural estimated per capita consumption	14750.055	3331.735	4905.195	122849.258	35729
Poverty rate	0.420	0.150	0.000	1.000	35729
Total count of households	464.937	621.126	1.000	9006.000	35736
Share of individuals primary school or above	0.371	0.176	0.000	1.000	35736
Share of population secondary school or above	0.126	0.094	0.000	1.000	35736
Land owner share	0.395	0.256	0.000	1.000	35736
Unirrigated land per household (in acres)	5.314	98.201	0.000	9068.889	35736
Per HH land with irrigation for two crops (acres)	6.246	95.197	0.000	9075.906	35736
Mean number of dwelling rooms per HH	2.259	0.734	0.425	8.000	35736
SC/ST share	0.211	0.208	0.000	1.000	35736
District HQ distance	32.964	19.027	0.250	190.000	33064
Block distance	10.135	7.139	0.000	222.000	33064
Community health center	0.001	0.033	0.000	1.000	33064
Post office	0.256	0.436	0.000	1.000	33064
Telephone	0.252	0.434	0.000	1.000	33064
Pucca road	0.647	0.478	0.000	1.000	33064
All-weather road	0.702	0.457	0.000	1.000	33064
Power Supply availability	0.634	0.482	0.000	1.000	33064
Power Supply Summer (April-Sept.)	2.844	2.992	0.000	22.000	33064
Power Supply Winter (Oct.-March)	3.265	3.389	0.000	22.000	33064
Area under Non-Agricultural Uses (in Hectares)	31.280	93.725	0.000	5690.200	33064

that the average hours of electricity supply are negatively correlated with the share of SC-ST population at feeder level. This is similar to Khanna and Rowe (2024) - they use the proportion of industrial consumers on the feeder in Delhi. I run the following 2SLS-IV specification:

$$Y_{i,t} = \alpha + \beta_1 S_{i,t} + X_i \delta + \epsilon_{i,t}$$

$$S_{i,t} = \beta_0 + \beta_1 u_i + X_i \gamma + \mu_{i,t}$$

where,

- $Y_{i,t}$: development outcomes, including education, health, labour force participation, etc for village i in year t.
- $S_{i,t}$: average daily hours of domestic electricity supply for village i in year t
- u_i: proportion of SC/ST population on the feeder
- X_i: vector of controls, including total households, distance from district HQ, schools, etc.

Figure 5 shows a visual representation of the treatment and control for the compliers. For descriptive comparisons, I define treatment and control villages based on the compliers implied

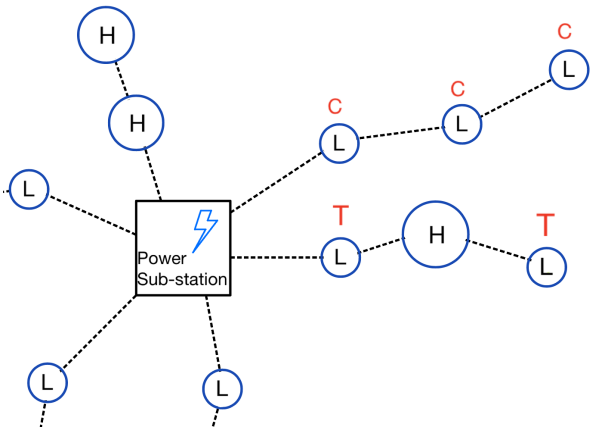


Figure 5: Identification strategy

by the IV:

- Treatment: SC-ST majority villages on upper-caste majority feeders
- Control: SC-ST majority villages on SC-ST majority feeders

As a complementary strategy, I also construct an income-based feeder instrument. This instrument is defined as the proportion of low-income villages connected to the feeder. The logic parallels the caste-based instrument: feeders serving a higher share of economically disadvantaged villages receive fewer supply hours, while low-income villages connected to relatively higher-income feeders receive greater supply than they otherwise would.

The income-based instrument serves two purposes. First, it provides an alternative source of feeder-level variation in electricity supply. Second, and central to this paper, it allows a substantive comparison between two forms of disadvantage. The caste-based instrument captures variation in supply generated by the social composition of the feeder, while the income-based instrument captures variation generated by the economic composition of the feeder. Comparing the two specifications formally, in Section 7 assesses whether electricity reliability produces similar development responses among socially marginalized and economically disadvantaged villages.

5.0.1 Identification assumptions

The empirical strategy relies on the caste composition of the electricity feeder as an instrument for electricity supply hours. For this strategy to identify the causal impact of electricity reliability on development outcomes, two main conditions must hold: relevance and the exclusion restriction.

Relevance

The first requirement is that the caste composition of the feeder affects electricity supply

hours. Electricity distribution in Bihar occurs through power substations that supply electricity to villages via feeders. Because supply decisions are implemented at the feeder level, all villages connected to the same feeder receive similar electricity supply schedules.

If electricity supply decisions are influenced by local social or political factors, feeders serving villages with different caste compositions may receive different amounts of electricity supply. In socially stratified rural settings, dominant caste groups often exercise greater influence over local institutions and public service provision. Previous work has shown that social composition can influence the allocation of public goods in India (Banerjee and Somanathan (2007); Asher et al. (2024)). In this context, feeders serving villages with a higher proportion of politically influential groups may receive more reliable electricity supply.

Consistent with this mechanism, preliminary evidence from the Socio-Economic and Caste Census (SECC) and feeder-level supply data suggests that average electricity supply hours are negatively correlated with the share of Scheduled Caste and Scheduled Tribe households connected to the feeder. This provides the first-stage variation required for the instrumental variable strategy. Table 2 reports the first-stage results, discussed further in Section 6.

Exclusion restriction

The second requirement is that feeder caste composition affects village-level development outcomes only through its effect on electricity supply hours. In other words, conditional on village characteristics and geographic controls, the caste composition of the feeder should not directly influence development outcomes in a village.

This assumption is plausible in the present setting for several reasons. First, villages do not choose the feeder to which they are connected. Feeder boundaries are determined by the physical layout of the electricity distribution network and the location of power substations. As a result, villages connected to the same feeder are typically determined by geographic proximity rather than by socioeconomic characteristics.

Second, the instrument is defined at the feeder level rather than at the village level. While caste composition within a village may influence local development outcomes directly, the caste composition of other villages connected to the same feeder should not directly affect outcomes in a given village except through its effect on electricity supply.

Third, the empirical specification includes a rich set of controls for village characteristics, including population size, baseline infrastructure, and distance to administrative centers. These controls help account for potential correlations between social composition and local development conditions.

Potential threats to identification

A potential concern is that feeder composition may be correlated with broader patterns of regional development or political influence. For example, feeders serving wealthier or more developed areas may both have a lower share of disadvantaged groups and receive more reliable electricity supply. If such factors are correlated with development outcomes independently of electricity supply, the exclusion restriction may be violated.

To address this concern, the empirical analysis includes geographic and infrastructure controls and focuses on comparisons between villages with similar characteristics that are connected to feeders with different social compositions. In addition, robustness checks will examine alternative specifications and control sets to ensure that the results are not driven by underlying regional differences.

Under these assumptions, variation in feeder caste composition provides a source of quasi-experimental variation in electricity supply hours that can be used to identify the causal impact of electricity reliability on village-level development outcomes.

6 Reliability has real productive effects (building block)

[1] Before turning to the paper's central comparison, this section establishes that electricity reliability has real, interpretable productive effects in the pooled sample, the gains that Section 7 will then show are distributed unequally.

Figure 6 shows the balance between treatment and control villages on a number of standardized baseline parameters, controlling for the distance from the district and block HQ. The treatment and control here are defined based on the 'compliers' of the IV. SC-ST majority villages on upper caste feeders have been defined as treated, while SC-ST majority villages on lower caste feeders have been defined as controls. The treatment villages have a slightly lower share of enterprise ownership at baseline but seem to be fairly balanced on other parameters.

6.1 First-stage

The validity of the instrumental variable strategy requires that the caste composition of the electricity feeder be strongly correlated with electricity supply hours. To examine this relationship, I estimate the first-stage regression of predicted electricity supply hours on the share of Scheduled Caste and Scheduled Tribe (SC–ST) population connected to the feeder serving each village.

Table 2 presents the results of this first-stage regression. The coefficient on the SC–ST share

[1] All the figures use standardized variables, and tables use non-standardize for ease of comparison and interpretation, respectively

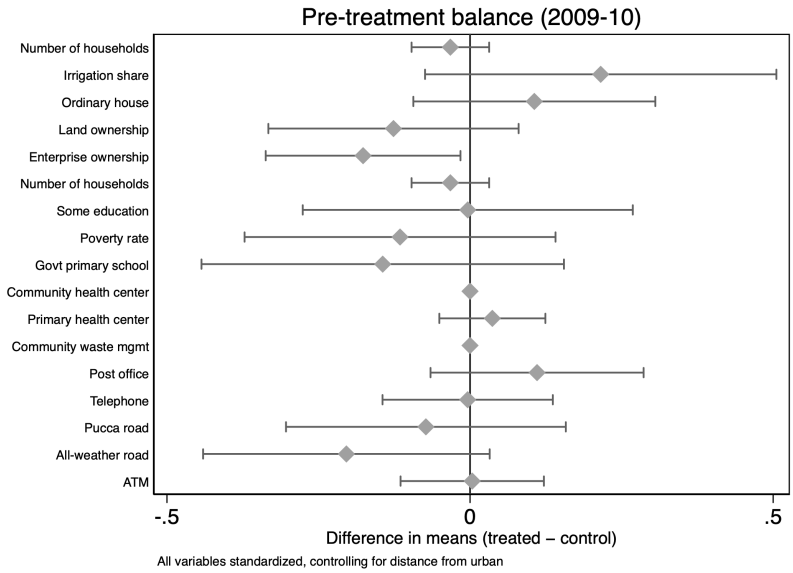


Figure 6: Balance between treatment and control

of the feeder is negative and statistically significant at the 1 percent level. The estimate implies that feeders with a higher proportion of SC–ST households receive fewer hours of electricity supply on average. This relationship is consistent with the hypothesis that electricity supply decisions may be influenced by the social composition of the villages served by the feeder.

The regression also includes a set of village-level controls to account for potential confounding factors. These controls include total population, education levels, land ownership share, availability of a pucca road, and distance from district and block headquarters. Including these controls helps ensure that the estimated relationship between feeder caste composition and electricity supply is not driven by differences in village size, infrastructure, or geographic accessibility.

The first-stage relationship is statistically strong. The F-statistic for the excluded instrument exceeds the conventional threshold of 10 used to assess weak instruments, indicating that the instrument has substantial explanatory power for electricity supply hours. This suggests that the caste composition of the feeder provides a meaningful source of variation in electricity reliability across villages.

The negative relationship between SC–ST share and electricity supply is consistent with the broader political economy literature documenting disparities in the allocation of public goods across social groups in India. Previous work has shown that socially disadvantaged communities often receive lower levels of public infrastructure and services. In this context, the results suggest that feeders serving villages with a higher proportion of marginalized groups may receive lower supply priority when electricity is rationed.

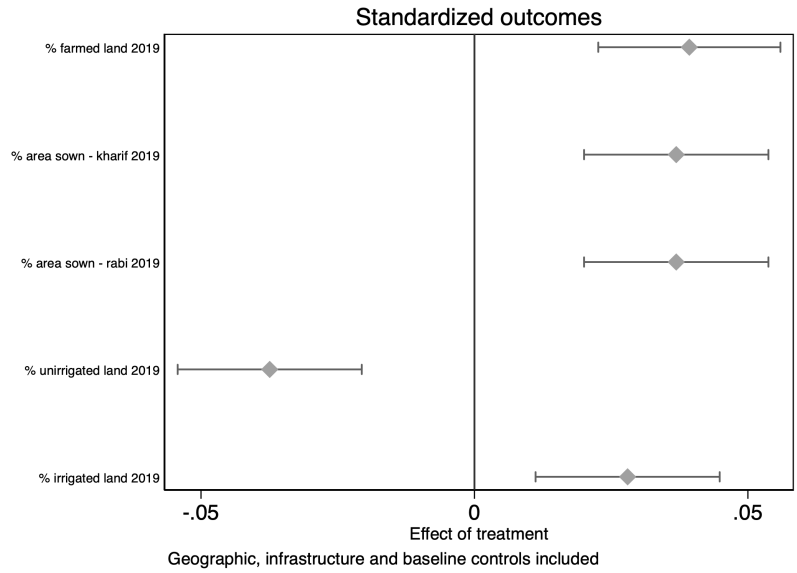


Figure 7: Farm activities - IV results

Note: Coefficients represent second-stage IV estimates where electricity supply hours are instrumented using the caste composition of the feeder. Outcomes are standardized. All regressions include geographic, infrastructure, and baseline village controls. Full regression results are reported in Appendix Table B1.

Taken together, these findings support the relevance condition for the instrumental variable strategy. Variation in feeder-level caste composition generates systematic differences in electricity supply hours across villages, providing the first-stage variation necessary to estimate the causal impact of electricity reliability on development outcomes.

Table 2: First-stage relationship between feeder caste/income composition and electricity supply

	(1)	(2)
Predicted hours	SC-ST share on feeder	Low-income share on feeder
Coefficient	-2.718***	-1.520***
	(0.225)	(0.123)
F-stat	555.6	589.7
Observations	33,062	33,062
Standard errors in parentheses		
*** p<0.01, ** p<0.05, * p<0.1		
Geographic, infrastructure and baseline controls included		

6.2 Farm activities

Figure 7 presents the estimated effects of electricity supply on agricultural activity using the instrumental variable strategy described in the previous section. The coefficients shown correspond to the second-stage estimates from the IV specification, where electricity supply hours are instrumented using the caste composition of the feeder. The full regression results for these specifications are reported in Appendix Table B1. All regressions include geographic,

infrastructure, and baseline village controls.

The results suggest that more reliable electricity supply increases agricultural activity. In particular, villages predicted to receive higher electricity supply exhibit a larger share of irrigated land and a correspondingly smaller share of unirrigated land. Consistent with this pattern, the estimates also indicate increases in the proportion of land farmed as well as the share of area sown during both the kharif and rabi seasons. Although the magnitudes are modest, the coefficients are consistently positive for most measures of agricultural activity.

These findings are consistent with the role of electricity in agricultural production in rural Bihar. Irrigation in the state relies heavily on groundwater extraction using electrically powered pump sets. Reliable electricity supply enables farmers to operate irrigation pumps more frequently and at lower cost, increasing the share of land that can be irrigated and cultivated. In contrast, villages facing limited electricity supply are more likely to rely on rain-fed agriculture, which restricts the extent of cultivated land and increases the share of land left unirrigated.

Electricity availability is particularly important during the rabi season, when rainfall is limited and irrigation becomes essential for crop production. Improvements in electricity reliability can therefore increase the area sown in both kharif and rabi seasons by allowing farmers to supplement rainfall with groundwater irrigation. This mechanism is consistent with the broader literature highlighting the importance of electricity for irrigation and agricultural productivity in India.

Taken together, the IV estimates suggest that more reliable electricity supply increases agricultural activity primarily through improvements in irrigation capacity. While the estimated effects are relatively small, they are consistent across several related agricultural outcomes and align with the institutional features of electricity use in rural agriculture. This is Proposition 1 of the model (Section 3.2): the farm channel is local and should appear for all villages.

6.3 Non-farm activities

Figure 8 presents the estimated effects of electricity supply on non-farm activity and related economic indicators. As in the previous analysis, the coefficients correspond to the second-stage estimates from the instrumental variable specification in which electricity supply hours are instrumented using the caste composition of the feeder. All regressions include geographic, infrastructure, and baseline village controls. The full regression results corresponding to these estimates are reported in Appendix Table B1.

The results suggest that more reliable electricity supply is associated with modest changes in local economic activity. Villages predicted to receive greater electricity supply exhibit a higher

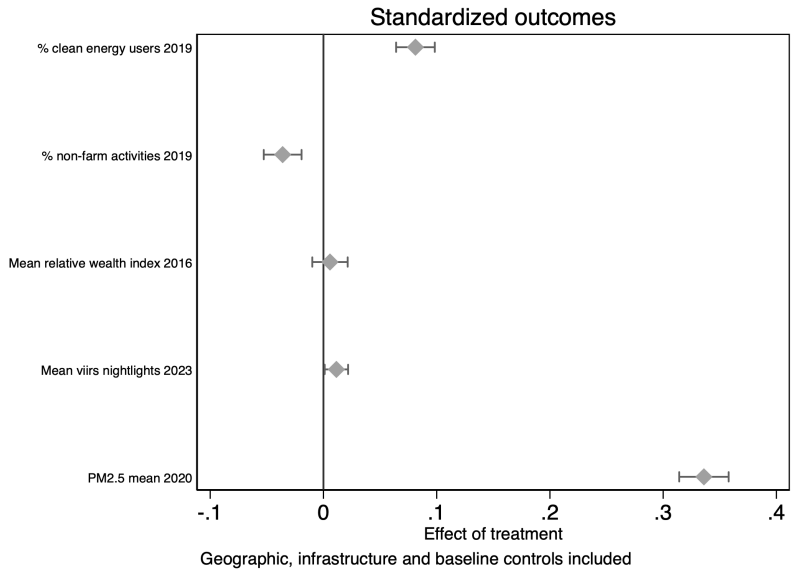


Figure 8: Non-farm activities - IV results

Note: Coefficients represent second-stage IV estimates where electricity supply hours are instrumented using the caste composition of the feeder. Outcomes are standardized. All regressions include geographic, infrastructure, and baseline village controls. Full regression results are reported in Appendix Table B1.

share of households using clean energy sources. This effect is consistent with the role of electricity access in facilitating the adoption of cleaner cooking and household energy technologies. Reliable electricity reduces reliance on traditional fuels such as biomass and kerosene and can complement the use of cleaner energy sources, including LPG or electric appliances.

The estimates also show a positive effect on nightlight intensity, which is commonly used as a proxy for economic activity. This pattern is consistent with the idea that more reliable electricity supply increases nighttime economic activity and household energy consumption, which in turn increases luminosity captured by satellite imagery.

In contrast, the coefficient for non-farm employment is negative, though relatively small in magnitude. One possible explanation is that improved electricity reliability may increase the productivity of agricultural activities, thereby reducing the need for households to diversify into non-farm employment. In rural Bihar, where agriculture remains the primary source of income for many households, improved irrigation and farm productivity can increase returns to farming and reduce the incentive to engage in non-farm activities.

The results also show a large positive coefficient for particulate matter (PM2.5). While this estimate should be interpreted with caution, it may reflect increased economic activity associated with higher electricity consumption and agricultural production, which can raise local pollution levels through increased mechanization, irrigation pumping, and related activities.

Overall, the IV estimates suggest that more reliable electricity supply is associated with in-

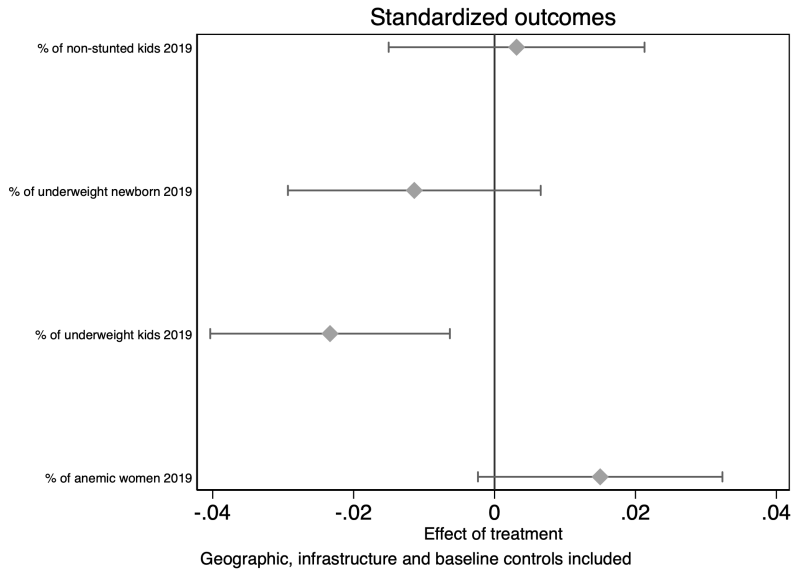


Figure 9: Health - IV results

Note: Coefficients represent second-stage IV estimates where electricity supply hours are instrumented using the caste composition of the feeder. Outcomes are standardized. All regressions include geographic, infrastructure, and baseline village controls. Full regression results are reported in Appendix Table B1.

creased household energy use and higher levels of nighttime economic activity, with more limited evidence of structural shifts toward non-farm employment. These results are broadly consistent with the structure of rural economies in Bihar, where electricity improvements primarily affect agricultural production and household energy consumption rather than immediately generating large-scale transitions into non-farm sectors.

6.4 Health

Figure 9 presents the estimated effects of electricity supply on several health-related outcomes. As in the previous analyses, the coefficients correspond to the second-stage estimates from the instrumental variable specification in which electricity supply hours are instrumented using the caste composition of the feeder.

The results provide limited evidence of improvements in health outcomes associated with more reliable electricity supply. The estimates for several indicators of child nutrition, including the share of non-stunted children and the share of underweight children, are relatively small and imprecisely estimated. While the coefficient on the proportion of non-stunted children is positive, the estimates for underweight children and underweight newborns are negative, though none of these effects appear to be statistically distinguishable from zero.

Similarly, the coefficient for the share of anemic women is positive but modest in magnitude. Taken together, these results suggest that improvements in electricity supply alone may not

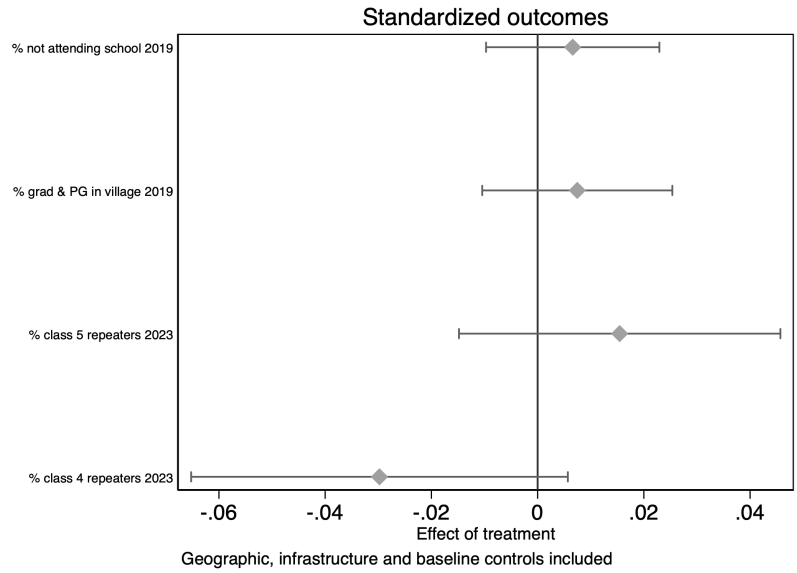


Figure 10: Education - IV results

Note: Coefficients represent second-stage IV estimates where electricity supply hours are instrumented using the caste composition of the feeder. Outcomes are standardized. All regressions include geographic, infrastructure, and baseline village controls. Full regression results are reported in Appendix Table B1.

translate into substantial changes in short-run health outcomes.

This pattern is consistent with the nature of the health production process in rural settings. While electricity can improve the functioning of health facilities, enable refrigeration for medicines and vaccines, and improve lighting for healthcare provision, many key determinants of health outcomes, such as nutrition, sanitation, access to healthcare services, and household income, depend on a broader set of factors. As a result, improvements in electricity reliability may have only limited direct effects on health indicators, particularly in the short run.

In addition, many of the health outcomes considered here reflect longer-term nutritional and demographic processes. Indicators such as child stunting and anemia are typically influenced by cumulative conditions over several years, including maternal health, dietary quality, and access to public health services. Consequently, changes in electricity supply may take longer to translate into measurable improvements in these outcomes.

Overall, the IV estimates suggest that while electricity reliability may influence economic activity and agricultural production, its effects on health outcomes appear to be more limited in the short run.

6.5 Education

Figure 10 presents the estimated effects of electricity supply on education-related outcomes. The estimates provide mixed evidence on the relationship between electricity reliability and

education outcomes. The coefficients for the share of children not attending school and the share of graduates and postgraduates in the village are small and imprecisely estimated, suggesting limited immediate effects of electricity supply on these measures of educational attainment.

The estimates for grade repetition show somewhat stronger patterns. In particular, the coefficient for the share of students repeating class 4 is negative, indicating that villages predicted to receive more reliable electricity supply experience lower repetition rates at this grade level. This estimate is statistically significant at the 10 percent level. In contrast, the coefficient for class 5 repetition is positive but imprecisely estimated, and therefore does not provide clear evidence of an effect.

One potential mechanism linking electricity supply to educational outcomes is improved household lighting. Reliable electricity allows students to study after dark and may increase the time available for homework and learning activities. Electricity can also improve school infrastructure by enabling lighting and the use of electronic equipment. However, the relatively modest effects observed here suggest that improvements in electricity reliability alone may not be sufficient to generate large short-run changes in educational attainment.

This pattern is consistent with the broader literature on electrification and education, which finds that while electricity can improve study conditions and school environments, educational outcomes are also shaped by a wide range of other factors including school quality, teacher availability, household income, and parental education. As a result, improvements in electricity supply may translate into gradual improvements in learning outcomes rather than immediate changes in enrollment or educational attainment.

Overall, the IV estimates suggest that electricity reliability may contribute to improvements in certain dimensions of educational performance, such as grade progression, while having more limited effects on broader educational outcomes in the short run.

6.6 Mechanisms: Irrigation and agricultural production

The results presented above suggest that the primary channel through which electricity reliability affects economic activity in rural Bihar is agricultural production. In particular, the IV estimates indicate increases in irrigated land and cropping intensity in villages predicted to receive more reliable electricity supply. This section discusses the mechanisms underlying these effects.

Electricity and Irrigation

In much of rural India, irrigation relies heavily on groundwater extraction using electric pump sets. Electricity is therefore a critical input in agricultural production, particularly in regions

where rainfall is seasonal and unreliable. Bihar is characterized by a monsoon-dependent agricultural cycle, with rainfall concentrated during the kharif season and relatively dry conditions during the rabi season. As a result, irrigation plays an especially important role in enabling farmers to cultivate crops outside the monsoon period.

Reliable electricity supply allows farmers to operate irrigation pumps more frequently and at lower cost. When electricity supply is intermittent or limited to a few hours per day, farmers may face constraints in irrigating fields at the appropriate time, which can reduce crop yields or limit the share of land that can be cultivated. In contrast, villages receiving more reliable electricity supply can irrigate land more consistently, increasing both the share of land that is irrigated and the total area under cultivation.

Consistent with this mechanism, the IV results show that villages predicted to receive more reliable electricity supply exhibit a higher share of irrigated land and a corresponding reduction in unirrigated land. The estimates also indicate increases in the proportion of land cultivated and in the share of land sown during both the kharif and rabi seasons.

Cropping Intensity and Agricultural Activity

Electricity reliability may also affect agricultural activity by enabling higher cropping intensity. In regions with reliable irrigation, farmers are more likely to cultivate multiple crops within a year. For example, fields that are cultivated during the monsoon season may be replanted during the winter season if irrigation water is available.

The results provide evidence consistent with this mechanism. Villages receiving more reliable electricity supply exhibit higher shares of land sown in both kharif and rabi seasons. This suggests that electricity reliability enables farmers to expand cultivation beyond rain-fed agriculture and adopt more intensive cropping patterns.

Higher cropping intensity can increase agricultural output and rural economic activity even without large changes in employment or sectoral composition. This may explain why the results show stronger effects on agricultural outcomes than on non-farm employment, and it anticipates the model's Proposition 2: when the non-farm margin is gated, electricity pulls effort further into agriculture.

Implications for Rural Economic Activity

The irrigation channel also helps explain several other results observed in the data. Increased irrigation and agricultural production may raise household income and electricity consumption, which could contribute to the higher nightlight intensity observed in villages receiving more reliable electricity supply. At the same time, higher agricultural productivity may reduce the need for households to diversify into non-farm employment, potentially explaining the negative

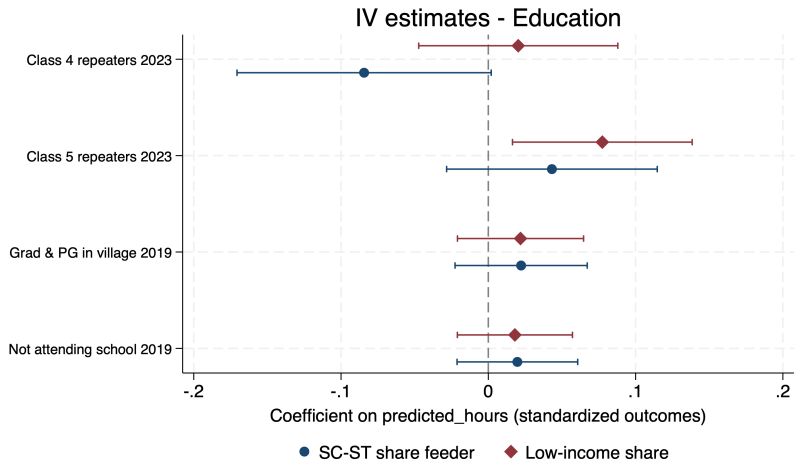


Figure 11: Education - caste vs income

Note: Coefficients represent second-stage IV estimates where electricity supply hours are instrumented using the caste or income composition of the feeder. Outcomes are standardized. All regressions include geographic, infrastructure, and baseline village controls.

or small effects observed for non-farm labor participation.

Taken together, these findings suggest that electricity reliability affects rural economic activity primarily through its role in enabling irrigation and increasing agricultural production. In regions such as Bihar, where agriculture remains the dominant economic activity, improvements in electricity reliability may therefore have their largest impacts through changes in agricultural practices rather than through immediate structural transformation toward non-farm sectors. This pooled picture, however, averages over a crucial distinction—the one the next section draws out.

7 Caste vs Income: Unequal Returns to Electricity Reliability

The preceding section establishes that electricity reliability raises productive outcomes in rural Bihar on average. But disadvantaged populations are not homogeneous, and as the model in Section 3.2 makes precise, the form of disadvantage determines whether reliability translates into broad-based gains or narrow agricultural ones. Rural development policy frequently treats caste and income disadvantage as interchangeable dimensions of deprivation. Yet income poverty primarily constrains material resources and productive capital (κ in the model), whereas caste disadvantage additionally restricts access to labour markets, social networks, credit, political influence, and occupational mobility (low A). Because electricity can help relax the former but not the latter, the two need not yield comparable development trajectories.

To examine this question, I compare two feeder-level instruments that capture distinct dimensions of disadvantage. The first uses the proportion of SC/ST villages on a feeder. The

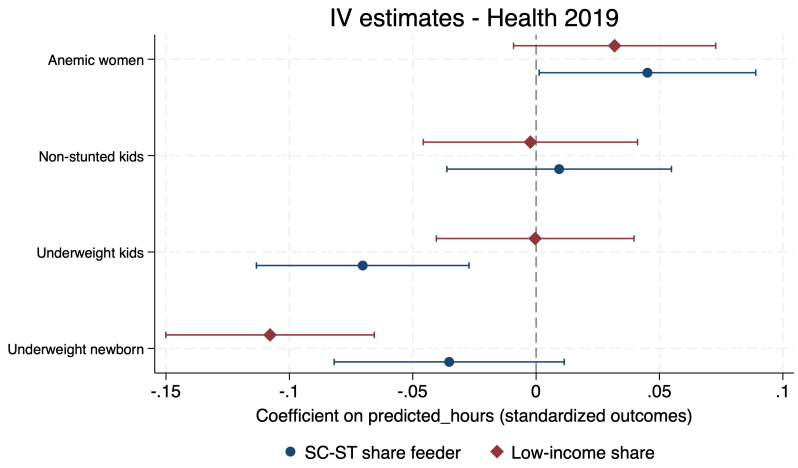


Figure 12: Health - caste vs income

Note: Coefficients represent second-stage IV estimates where electricity supply hours are instrumented using the caste or income composition of the feeder. Outcomes are standardized. All regressions include geographic, infrastructure, and baseline village controls.

second uses the proportion of low-income villages on a feeder. In both cases, feeders serving more disadvantaged villages receive lower electricity supply (Table 2). The institutional logic is therefore comparable across the two specifications, allowing a comparison of how electricity reliability operates across different forms of marginalization.

Importantly, the two instruments need not identify identical complier populations. Villages whose electricity supply responds to feeder caste composition may differ systematically from villages whose supply responds to feeder income composition. Rather than viewing this as a limitation, I interpret the comparison substantively. The exercise asks whether electricity reliability produces similar development responses when disadvantage is primarily economic versus when disadvantage is embedded within social hierarchy.

The results reveal important differences across the two specifications. Under the income-based instrument, improvements in electricity reliability generate broad-based productive gains. Higher electricity supply increases agricultural activity, strengthens non-farm outcomes, and leads to positive improvements in village wealth indicators. The pattern is consistent with electricity relaxing productive constraints in economically deprived settings (Proposition 4, income case): improved supply supports both farm and non-farm activity, suggesting that economically disadvantaged villages translate infrastructure improvements into broader local economic expansion.

The caste-based specification paints a different picture. As with the income IV, electricity reliability improves agricultural outcomes (fig. 13). However, these gains are accompanied by weaker non-farm outcomes and little evidence of wealth accumulation (fig. 14). Rather

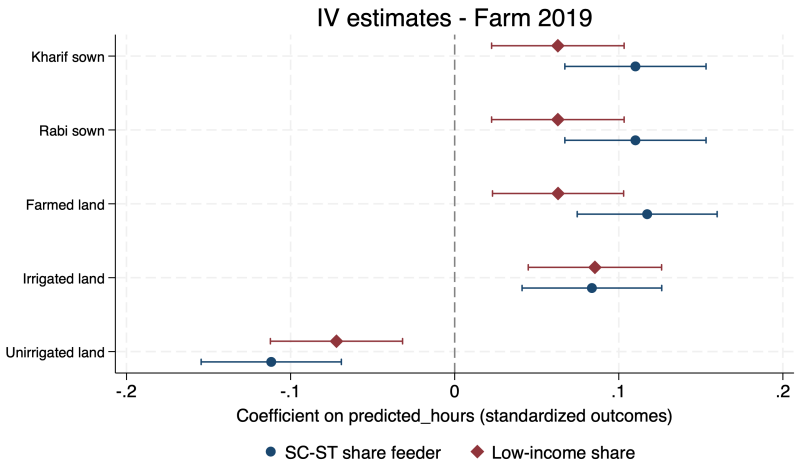


Figure 13: Farm - caste vs income

Note: Coefficients represent second-stage IV estimates where electricity supply hours are instrumented using the caste or income composition of the feeder. Outcomes are standardized. All regressions include geographic, infrastructure, and baseline village controls.

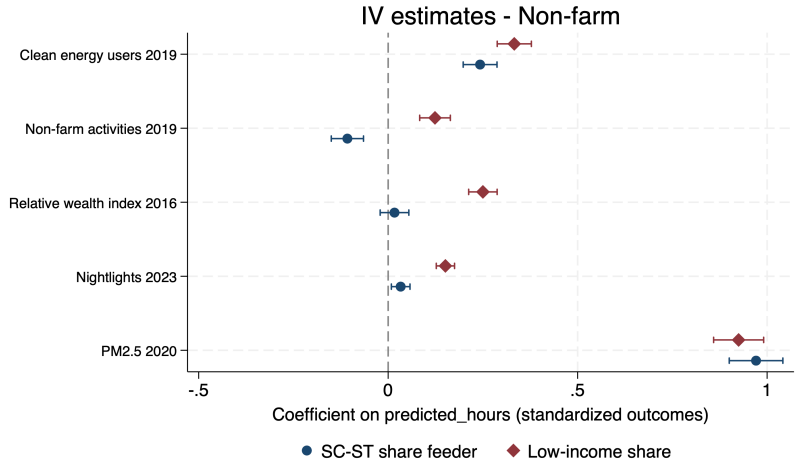


Figure 14: Non-farm - caste vs income

Note: Coefficients represent second-stage IV estimates where electricity supply hours are instrumented using the caste or income composition of the feeder. Outcomes are standardized. All regressions include geographic, infrastructure, and baseline village controls.

than broadening economic opportunity across sectors, electricity improvements among caste-marginalized villages appear to be associated with a reallocation toward agriculture and away from non-farm activity, exactly Proposition 2 of the model when $A < A^*$.

This divergence is substantively important. The model's interpretation is that electricity reliability alone is insufficient to generate structural transformation when complementary access A is unequally distributed. Non-farm expansion requires inputs beyond electricity: market connections, mobility, enterprise networks, credit, productive assets, and entry into occupations that may themselves be socially segmented. Socially marginalized groups face barriers along several of these margins. Consequently, the same improvement in reliability that enables diversification in low-income villages produces narrower agricultural adjustment in caste-disadvantaged settings.

The contrast also offers a broader lesson for infrastructure policy. Development gains from public infrastructure need not be distributionally neutral. Improvements in reliability can raise productivity without generating inclusive growth. If economically disadvantaged villages convert improved electricity into gains across agriculture, non-farm activity, and wealth, while caste-marginalized villages remain concentrated in agriculture, infrastructure improvements may contribute to divergent long-run development trajectories.

This does not imply that electricity reliability harms caste-marginalized communities. Agricultural improvements are meaningful welfare gains in their own right. Rather, infrastructure investments interact with existing social constraints in ways that shape who captures broader economic opportunities. Reliable electricity can relax productive constraints, but the returns depend on the social environment in which households operate.

These findings contribute to a literature on heterogeneous returns to infrastructure. Much existing evidence studies average treatment effects or income heterogeneity; the results here suggest that *social* disadvantage, distinct from poverty, fundamentally alters how infrastructure translates into development. Policies that improve reliability may therefore require complementary interventions addressing social exclusion, labour-market segmentation, access to productive assets, or barriers to non-farm participation if their benefits are to be distributed evenly.

8 Conclusion

This paper asks whether a single infrastructure improvement, greater electricity reliability, delivers comparable development gains across different forms of disadvantage in rural India. Exploiting feeder-level supply allocation in Bihar and using the caste and income composition of the feeder as instruments for village supply hours, the central finding is that it does not.

Economically disadvantaged villages translate improved reliability into broad-based gains spanning farm activity, non-farm activity, and wealth. Caste-disadvantaged villages, subject to the same supply mechanism, see gains concentrated in agriculture, with weaker non-farm outcomes and no comparable wealth gains. A simple model accounts for the pattern through a single distinction: electricity relaxes the material constraint that binds the income-poor but not the social constraint - exclusion from credit, markets, and non-farm occupations, that binds the caste-marginalized.

The supporting results explain why agriculture is the common channel. Villages predicted to receive greater supply exhibit higher irrigated-land shares and greater cropping intensity in both kharif and rabi seasons, consistent with the central role of electricity in groundwater irrigation in Bihar. Beyond agriculture, the pooled estimates show modest increases in clean-energy adoption and economic activity, limited short-run evidence of structural transformation toward non-farm employment, small and insignificant health effects, and modest education effects (lower primary-grade repetition). These dimensions depend on a broad set of factors and respond more slowly than agricultural margins.

Taken together, the results highlight the importance of reliability, not access alone, in shaping rural development, and the role of social and political factors in allocating it: feeder-level decisions that track social composition can reinforce existing inequalities. The policy implication is not that reliability is unimportant, it raises agricultural productivity and rural livelihoods, but that infrastructure alone may be insufficient to reduce inequality when complementary opportunities remain socially constrained. Where caste-marginalized communities face barriers to credit, markets, non-farm employment, or productive assets, reliability improvements may raise agricultural productivity without enabling structural transformation, and could widen long-run gaps unless paired with policies addressing those social and institutional barriers. Strengthening transparency and monitoring in feeder-level distribution may also help ensure more equitable provision.

Several extensions follow. Better administrative data on supply hours would sharpen the reliability measure; longer panels would reveal whether reliability eventually translates into structural change; and further work could examine how political incentives shape supply allocation across communities. By focusing on marginalized communities and exploiting the structure of distribution networks, this study provides new evidence that the quality of infrastructure, and its interaction with social inequality, matters for who develops.

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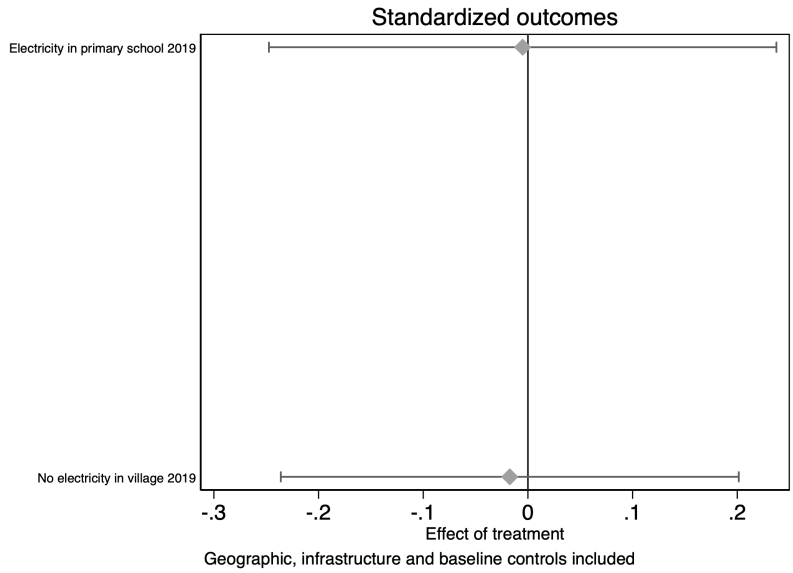


Figure A1: Electrification in 2019

A Appendix 1

A.1 Results using caste based treatment and control assignment

As a robustness check, I have also run an analysis on 'compliers'. Based on the IV, the treatment and control groups will be:

- Treatment: SC-ST majority villages on upper-caste majority feeders
- Control: SC-ST majority villages on SC-ST majority feeders

The regression I run for the analyses in the next section is:

$$Y_i = \alpha + \beta_1 T_i + X_i \delta + \epsilon_i$$

I've included geographic, infrastructure, and village characteristic controls.

The first thing to check is whether they are similarly electrified after 2018. Figure A1 shows this based on two measures of electrification that I have in the data. There seems to be no difference in electrification levels.

Figure A2 shows the effects on employment and education. There is a negative effect on non-farm activities, although not significant. Coefficient on farm activities, proportion of graduates and post-graduates, and proportion of kids not attending school are all positive and insignificant.

Figure A3 shows the effect on health related outcomes. The treatment villages are using more clean energy. But there doesn't seem to be any effect on health outcomes. The coefficients

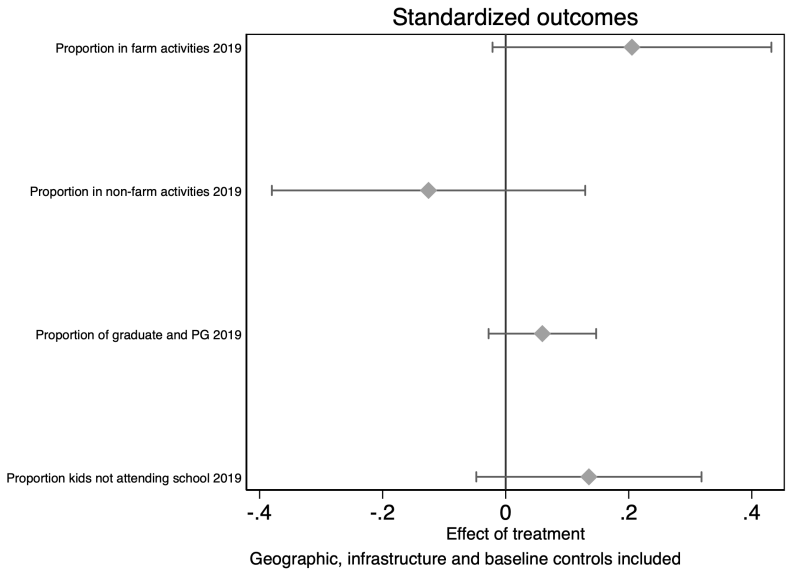


Figure A2: Effect on employment and education

on proportion of anemic women, underweight kids, and non-stunted kids are all negative but insignificant. Figure A4 shows that treatment villages have a higher mean nightlights in 2023, and a higher PM2.5 level in 2020. There is a positive coefficient on wealth index in 2016 but it is not significant.

The results in this section are noisier but align with the IV results in terms of the direction of the effect.

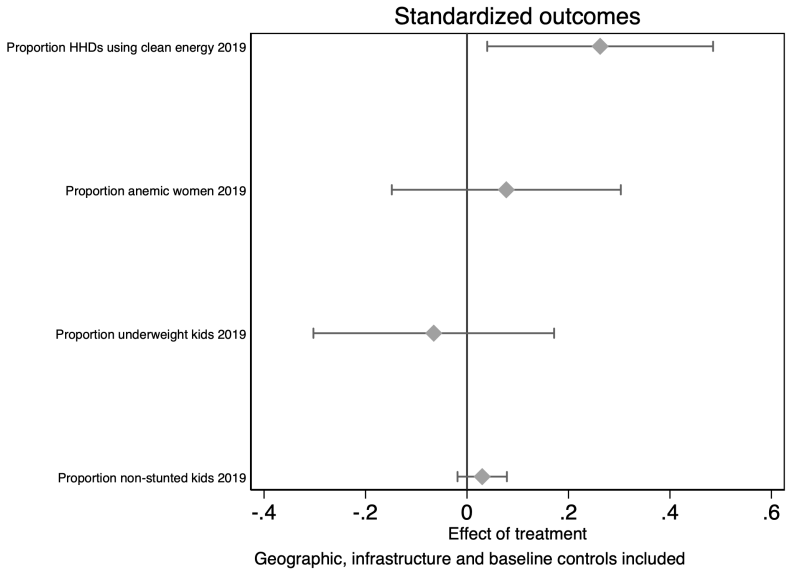


Figure A3: Effect on health

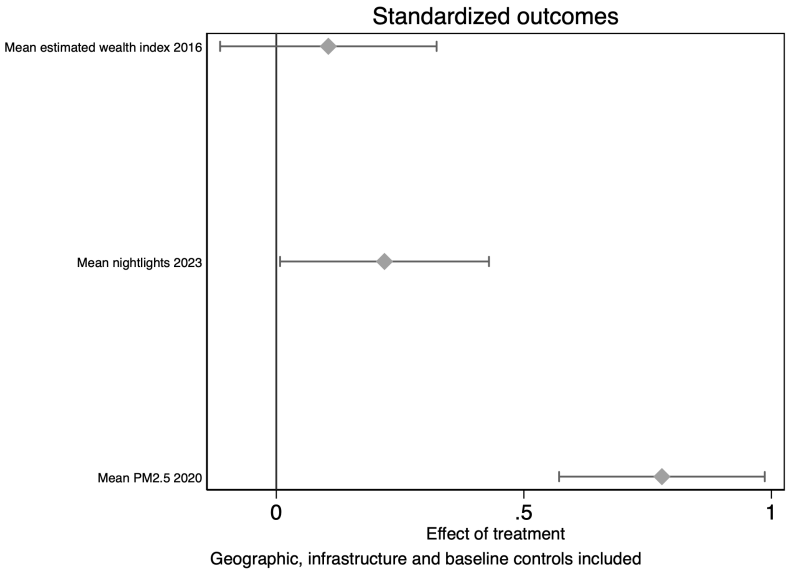


Figure A4: Effect on wealth and nightlights

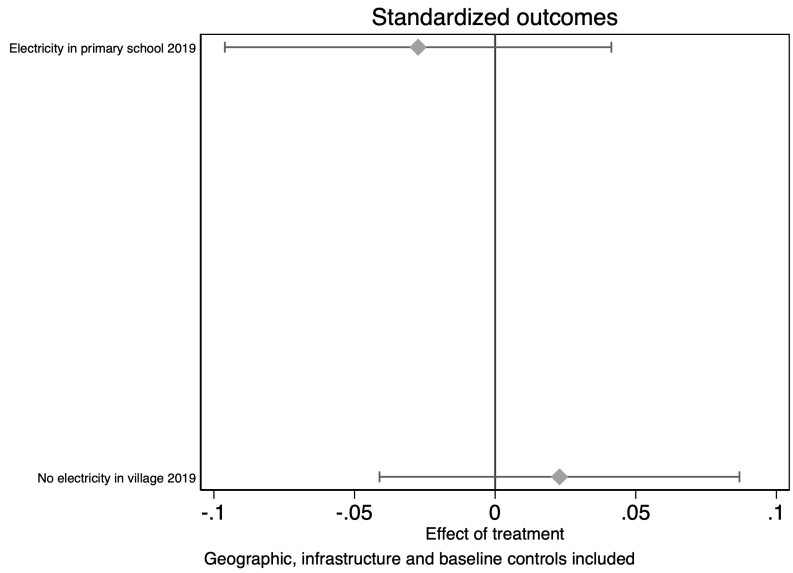


Figure A5: Electrification in 2019

A.2 Results using income based treatment and control assignment

The following results use income composition of the village with respect to that of the feeder in assigning treatment and control. Villages in the bottom income quartile on feeders with high proportion of high-income-quartile villages are classified as treatment. And villages in the bottom income quartile on feeders with low proportion of high-income-quartile villages are classified as control.

The first thing to check is whether they are similarly electrified after 2018. Figure A5 shows this based on two measures of electrification that I have in the data. There seems to be no difference in electrification levels.

Figure A6 shows the effects on employment and education. There is a positive effect on non-farm activities. Coefficient on farm activities, proportion of graduates and post-graduates, and proportion of kids not attending school are all insignificant.

Figure A7 shows the effect on health related outcomes. The treatment villages are using more clean energy. But there doesn't seem to be any effect on health outcomes. The coefficients on proportion of anemic women, underweight kids, and non-stunted kids are all negative but insignificant. Figure A8 shows that treatment villages have a higher mean nightlights in 2023, and a higher PM2.5 level in 2020. There is a positive coefficient on wealth index in 2016 but it is not significant.

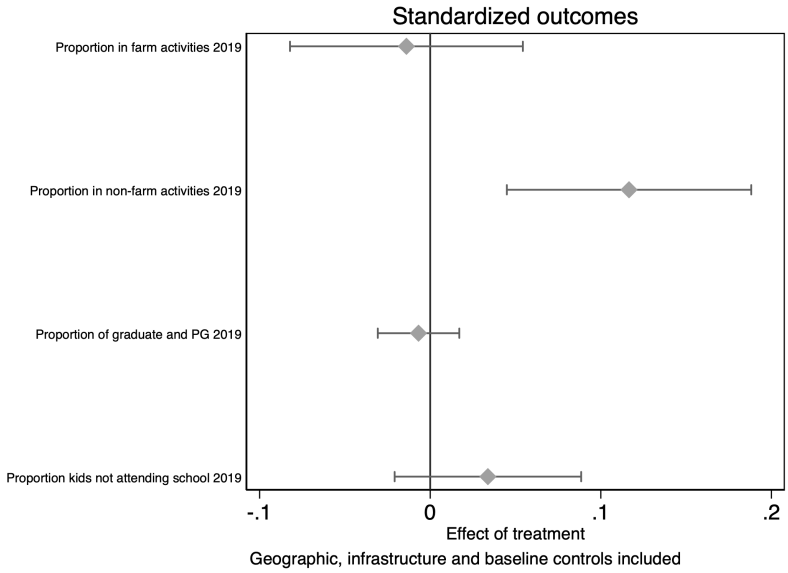


Figure A6: Effect on employment and education

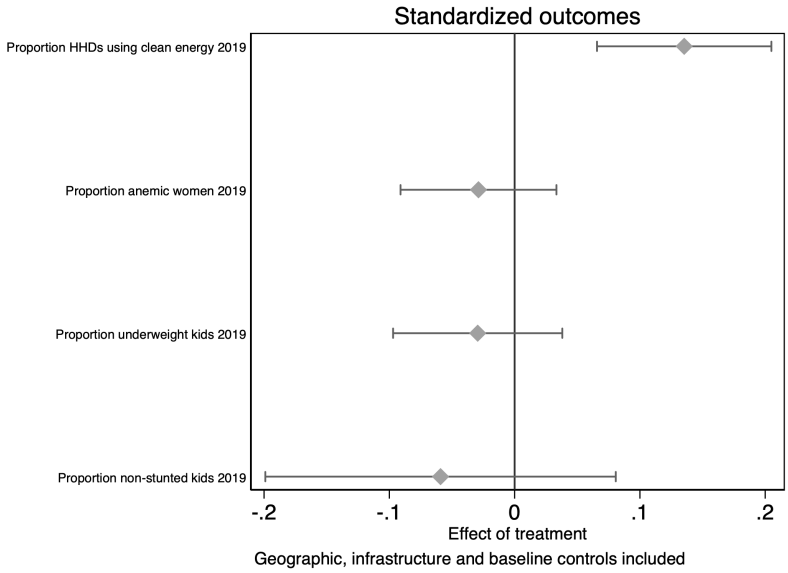


Figure A7: Effect on health

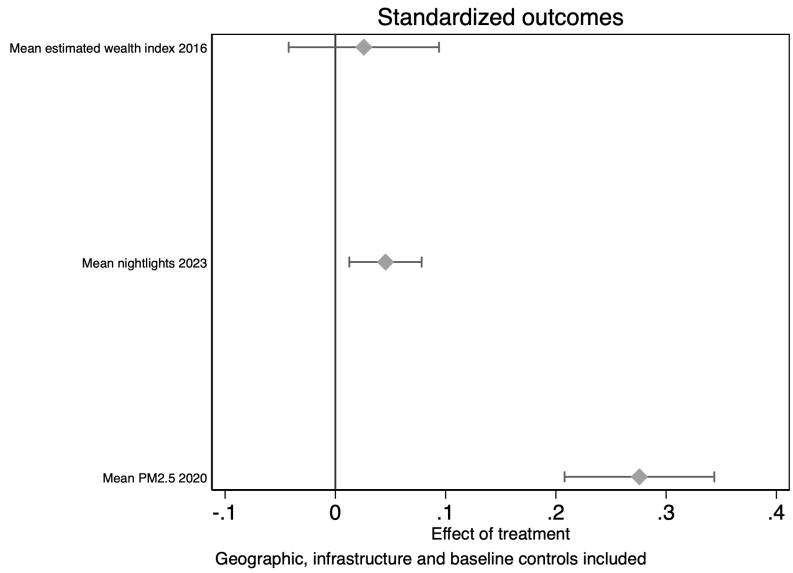


Figure A8: Effect on wealth and nightlights

B Appendix 2

B.1 Tables

Table B1: Second stage IV results (caste instrument)

	(1)	(2)	(3)	(4)	(5)
Outcomes	Prop in farm activities 2019	Prop sown in Kharif 2019	Prop sown in Rabi 2019	Prop unirrigated 2019	Prop irrigated 2019
Predicted supply	0.0106***	0.0106***	0.0106***	-0.00813***	0.00909***
	(0.00229)	(0.00246)	(0.00246)	(0.00186)	(0.00279)
Observations	32,676	32,676	32,676	32,676	32,676
Outcomes	Proportion HHDs using clean energy 2019	Proportion in non-farm activities 2019	Mean estimated wealth index 2016	Mean nightlights 2023	Mean PM2.5 2020
Predicted supply	0.0246***	-0.00609***	0.00186	0.0246**	4.608***
	(0.00264)	(0.00144)	(0.00256)	(0.0113)	(0.153)
Observations	32,676	32,676	33,004	33,062	33,062
Outcomes	Proportion kids not attending school 2019	Proportion of graduate and PG 2019	2023 prop class 5 repeaters	2023 prop class 4 repeaters	
Predicted supply	0.00264	0.00107	0.0340	-0.0519*	
	(0.00333)	(0.00131)	(0.0340)	(0.0316)	
Observations	32,676	32,676	10,110	9,470	
Outcomes	Proportion non-stunted kids 2019	prop_underw_newb2019	Proportion underweight kids 2019	Proportion anemic women 2019	
Predicted supply	0.00236	-0.00390	-0.00493***	0.00257*	
	(0.00700)	(0.00313)	(0.00183)	(0.00151)	
Observations	32,676	28,090	31,563	31,563	
Controls	yes	yes	yes	yes	yes

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

C Appendix 3

D Methodology

D.1 Objective

The objective of this study is to estimate **reliable electricity supply hours** for villages in Bihar using remotely sensed nightlight data combined with socio-economic characteristics. Measuring electricity reliability is difficult because consistent, high-frequency ground-level data is often lacking, especially in developing countries, where official records tend to be incomplete or unavailable Dugoua et al. (2022); Min et al. (2017). To address this limitation, the model is designed to learn a relationship between nightlight intensity, socio-economic characteristics, and electricity reliability, which can then be generalized across all villages in the state.

D.2 Dataset Construction

The first step involved constructing a training dataset to predict **reliable electricity supply hours** for villages in Bihar. This required integrating three main data sources:

- **Prayas Dataset (Ground Truth)**
  Source: https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/CLLZZM
- **Nightlight Data (VIIRS)**
  Source: https://developers.google.com/earth-engine/datasets/catalog/NOAA_VIIRS_DNB_MONTHLY_V1_VCMCFG
- **Socio-Economic and Caste Census (SECC)**
  Source: https://docs.devdatalab.org/SHRUG-Metadata/Socio-Economic%20and%20Caste%20Census%20%282012%29/Tables/secc-mord-rural-metadata/

D.2.1 Ground Truth Data Processing (Prayas Dataset)

The Prayas dataset provides **minute-level voltage readings** for multiple locations in Bihar over the period August 2015 to December 2019.

- Data available for **16 locations across 3 districts**: Jamui, Saharsa, and Rohtas.
- Locations were mapped to **SHRUG village IDs**:
  - Villages were directly matched

  - Localities were manually mapped to villages using **Google search**

- In cases where multiple localities mapped to a single village, values were **aggregated using averages**

This resulted in **12 unique villages**.

Defining Reliable Electricity

Reliable (normal) voltage was defined as:

205V - 270V

- For each hour, the number of minutes within the defined voltage range was computed. These were then aggregated across all hours in a day to obtain total daily normal minutes.
- The total was converted into **reliable hours** by dividing by 60.
- Monthly reliable electricity was calculated as the average of hourly reliable electricity over a month

Data Quality Issues and Handling

The dataset had several issues:

- Missing hourly observations
- Mislabelled hours (e.g., all hours recorded as 0)
- Duplicate entries

To address these, multiple strategies were adopted:

Missing Data Handling

- Created datasets with:
  - No missing hours
  - Missing hours ≤ 12
  - Missing hours of any size

Mislabelled Data Handling

- Included rows with:
  - 24 entries (assumed correct but mislabelled)
  - 48 entries with 24 duplicate entries (filtered accordingly)

Imputation Strategies

Missing values were imputed using both **mean** and **median**, with two approaches:

1. Location–Month–Hour based aggregation
2. Location–Hour based aggregation across time

The location–hour based imputation approach was motivated by the temporal structure of electricity supply, where availability tends to follow consistent hourly patterns within a location Dugoua et al. (2022).

Final Ground Truth Datasets

Combining all preprocessing and imputation strategies resulted in:

```
10 different ground truth datasets
```

Each representing different assumptions about data quality and imputation.

D.2.2 Nightlight Feature Construction

Night-time light intensity is widely used as a proxy for electricity access and reliability, since areas with more reliable electricity generally display higher and more stable levels of luminosity Dugoua et al. (2022); Min et al. (2017); Mann et al. (2016a).

- Used **SHRUG village polygons** to extract nightlight data from the VIIRS dataset available on Google Earth Engine NOAA and NASA (2024)
- Computed **monthly sum of radiance** for each village
- Applied log transformation:

```
log_nightlights = log(sum of radiance)
```

D.2.3 Socio-Economic Features (SECC)

Village-level SECC data was merged with the ground truth dataset, including:

- Population and household counts
- Literacy
- Caste composition
- Employment characteristics

D.2.4 Feature Sets

Two different feature sets were constructed:

Feature Set A (Absolute Variables)

- Log nightlights
- Area
- Log population
- Log households
- SC population
- ST population
- Literate population

Feature Set A was inspired by prior work that uses nightlight intensity along with demographic and structural village characteristics, such as population, area, and literacy, to predict hours of reliable electricity in the state of Uttar Pradesh Dugoua et al. (2022).

Feature Set B (Share-Based Variables)

- Log nightlights
- SC share
- ST share

- Literacy rate
- Worker rate
- Marginal worker share
- Agricultural share
- Household density
- Urban/Rural indicator

Feature Set B was constructed using normalized and share-based variables to capture relative socio-economic characteristics of villages, including caste composition, literacy, employment structure, and population density. This formulation focuses on proportional relationships rather than absolute levels.

D.2.5 Final Training Datasets

Combining:

- 10 ground truth datasets
- 2 feature sets

resulted in:

```
20 distinct modelling datasets
```

D.2.6 Prediction Dataset

For statewide prediction:

- Nightlight and SECC features were compiled for **all villages in Bihar** for all months from 2012 to 2021
- These were aligned with the feature structure of the training datasets

D.2.7 Construction of Zero-Hour Electricity Observations

To expand the range of reliable electricity hours in the training data, we constructed an additional dataset consisting of villages with zero hours of electricity supply. This was done using information on village electrification timelines. For each selected village, all observations prior to its electrification year were assigned zero reliable hours. Corresponding nightlight and

SECC features were also obtained for these villages. The resulting dataset was then merged with the original training data.

The data construction process involved extensive cleaning, imputation, and feature engineering across multiple datasets. Multiple dataset variants were created to assess robustness to data quality assumptions, forming the foundation for subsequent modelling and prediction.

D.3 Modelling Approach and Experiments

Following recent literature, machine learning methods such as decision trees and random forests are well suited for modelling electricity reliability due to their ability to capture nonlinear relationships between nightlight intensity and socio-economic characteristics Dugoua et al. (2022); Mann et al. (2016a).

The modelling process was conducted iteratively, with successive refinements to the dataset, feature set, and validation strategy. Each stage aimed to address limitations identified in the previous step.

D.3.1 Baseline Model

In the initial stage, baseline models were developed using the constructed datasets without incorporating zero-hour observations. The objective of this stage was to establish a reference performance for subsequent model improvements.

Two tree-based models were implemented:

- **Decision Tree Regressor**
- **Random Forest Regressor**

Both models were trained using two feature sets derived from nightlight and socio-economic data.

Model Specification

For the Random Forest model:

- Number of trees: 200
- Maximum depth: tuned over the range 3–10
- Minimum samples per leaf: 5
- Feature selection: square root of total features

For the Decision Tree model:

- Maximum depth: tuned over the range 3–15
- Minimum samples per leaf: 5

Validation Strategy

Model performance was evaluated using **5-fold cross-validation**. To reduce variability arising from random data splits, the cross-validation process was repeated multiple times with different random seeds.

The primary evaluation metric used was **Mean Absolute Error (MAE)**.

Hyperparameter Tuning

For both models, the optimal tree depth was selected based on the lowest average cross-validation MAE across all folds and repetitions.

Uncertainty Estimation (Bootstrapping)

For the Decision Tree model, prediction uncertainty was assessed using a bootstrap approach.

- Multiple bootstrap samples were generated by sampling the training data with replacement
- A Decision Tree model was trained on each bootstrap sample
- Predictions were generated for the full dataset using each model
- The standard deviation of predictions across bootstrap samples was computed

The average standard deviation across observations was used as a measure of prediction uncertainty and model stability.

D.3.2 Incorporating Zero-Hour Observations

The baseline model was extended by incorporating villages with zero electricity supply to better capture low-electrification conditions, which were underrepresented in the original dataset.

Experimental Setup

To assess the impact of zero-hour observations, experiments were conducted using different samples of zero-hour villages:

- A smaller sample consisting of 5 villages

- A larger sample consisting of 8 villages

These samples were selected to evaluate the sensitivity of model performance to the number of zero-hour observations included.

Both Decision Tree and Random Forest models were trained on the augmented datasets using the same feature sets and validation strategy as in the baseline model.

Evaluation Strategy

Model performance was evaluated using repeated 5-fold cross-validation, with Mean Absolute Error (MAE) as the primary metric.

In addition, out-of-fold (OOF) predictions were generated to assess model performance on unseen data.

Effect of Zero-Hour Observations

Including zero-hour observations introduced a broader range of electricity supply values, particularly improving representation of low-electrification conditions.

However, this also altered the distribution of the training data, increasing the proportion of low or zero values. As a result, the model exhibited a tendency to predict lower electricity supply levels, particularly in earlier periods.

The comparison between different sample sizes (5 vs 8 villages) provided insights into the sensitivity of model performance to the extent of zero-hour data inclusion.

Application to Bihar

Given the relatively small size of the training dataset (approximately 300 observations), models with lower values of the minimum samples per leaf parameter tended to fit the training data more closely.

However, when applied to the full Bihar dataset (approximately 38,000 villages), such models exhibited higher variability in predictions, indicating potential overfitting.

To address this, the minimum samples per leaf parameter was increased to produce smoother and more stable predictions at scale. This adjustment reflects a trade-off between model fit on the training data and generalization performance on a much larger prediction dataset.

Final model selection was therefore informed by both cross-validation performance and the stability of predictions across Bihar.

The trained models were then used to generate predictions for all villages in Bihar, using statewide nightlight and socio-economic data.

D.3.3 Temporal Modelling and Data Refinement

In this stage, the modelling approach was refined to explicitly account for temporal dynamics and improve the quality of the training data.

Temporal Validation and Feature Enhancement

To address temporal dependence in electricity supply, the validation strategy was updated to use **TimeSeriesSplit**, ensuring that models were trained on earlier periods and evaluated on later periods. Additionally, a time variable *year_ index* was introduced to capture the upward trend in electrification over time.

Data Refinement

It was observed that post-2018 data contained observations for only a single village, leading to potential imbalance. These observations were therefore excluded to ensure consistency in the training dataset.

Zero-Hour Village Selection Strategies

The construction of zero-hour observations was further refined to reduce bias and improve comparability with observed data.

Multiple strategies were explored:

- **Temporal filtering:** Zero-hour observations were restricted to periods closer to the electrification threshold (e.g., post-2013 or post-2014), rather than including very early periods. This was done to reduce the risk of the model learning a deterministic relationship between early years and zero electricity supply.
- **Random sampling:** Zero-hour villages were selected randomly under different temporal constraints.
- **Matched sampling:** Zero-hour villages were selected to have characteristics (such as population and area) comparable to non-zero villages.This was done to ensure that zero-hour and non-zero villages were comparable in size, since larger villages naturally have higher nightlight values, which could otherwise distort the relationship between nightlights and electricity supply.

These strategies resulted in multiple zero-hour datasets, enabling evaluation of how different assumptions affect model performance.

Model Choice

Based on baseline results, Random Forest models were found to outperform Decision Trees. Therefore, subsequent experiments in this stage focused primarily on Random Forest models.

D.3.4 Weighted Modelling to Address Bias from Zero-Hour Observations

The inclusion of zero-hour observations, constructed using electrification timelines, introduces potential bias in the training data. Since these observations are not directly measured but inferred, treating them equally can lead the model to overemphasize zero electricity supply, particularly in earlier years.

To mitigate this issue, we implemented a **sample weighting strategy**, assigning lower weights to pre-electrification observations while retaining full weight for observed data. Specifically:

- Observations prior to 2015 were assigned weights in the range of 0.1 to 0.7
- Observations from 2015 onwards were assigned a weight of 1

This approach allows the model to incorporate information from zero-hour observations while reducing their influence on the learned relationships.

Hyperparameter Tuning

A joint grid search was conducted over:

- Maximum tree depth (3–9)
- Sample weights for pre-2015 observations $(0.1, 0.2, 0.3, 0.5, 0.7)$

Model performance was evaluated using **Time Series Cross-Validation (TimeSeriesSplit)**, ensuring that training was performed on earlier periods and tested on later periods.

Results and Interpretation

The weighted model demonstrated improved predictive performance compared to the unweighted approach. This suggests that reducing the influence of artificially constructed zero-hour observations helps prevent the model from learning spurious relationships between time and electricity supply.

Overall, the weighting strategy provides a balanced approach, allowing zero-hour observations to inform the lower bound of electricity availability while preserving the integrity of observed data.

E Results

E.1 Overview of Modelling Progression

The modelling framework was developed iteratively to address key limitations in the data and improve the realism of predicted electricity supply. While both Decision Trees and Random Forest models were initially evaluated, Random Forest consistently demonstrated superior performance and stability. Therefore, the results presented focus on Random Forest models.

E.2 Model Performance

Table C1 presents the performance of models across different stages of the modelling pipeline.

Model	CV MAE	R^2
Baseline	2.79	0.71
Zero-hour	2.18	0.90
Time-series	3.41	0.77
Weighted	3.47	0.86

Table C1: Model performance across different stages

The inclusion of zero-hour observations leads to a reduction in CV MAE and a substantial increase in R^2, indicating improved fit due to the expanded range of the dependent variable. However, the transition to time-series cross-validation results in higher prediction error, reflecting a more realistic evaluation setting where the model must predict future observations without access to future information.

The weighted model shows a slight increase in error compared to the zero-hour model but maintains strong explanatory power. This suggests that weighting reduces overfitting to artificial observations and improves the robustness of predictions.

E.3 Prediction Analysis Across Bihar

To evaluate model behaviour beyond aggregate metrics, we analyze predicted electricity supply across all villages in Bihar.

E.3.1 Yearly Trends

Figure C1 presents the yearly mean predicted electricity supply.

The zero-hour model produces relatively flat trends, indicating limited ability to capture temporal improvements in electricity supply. In contrast, the time-series model exhibits a clear upward trajectory, reflecting the gradual expansion of electrification.

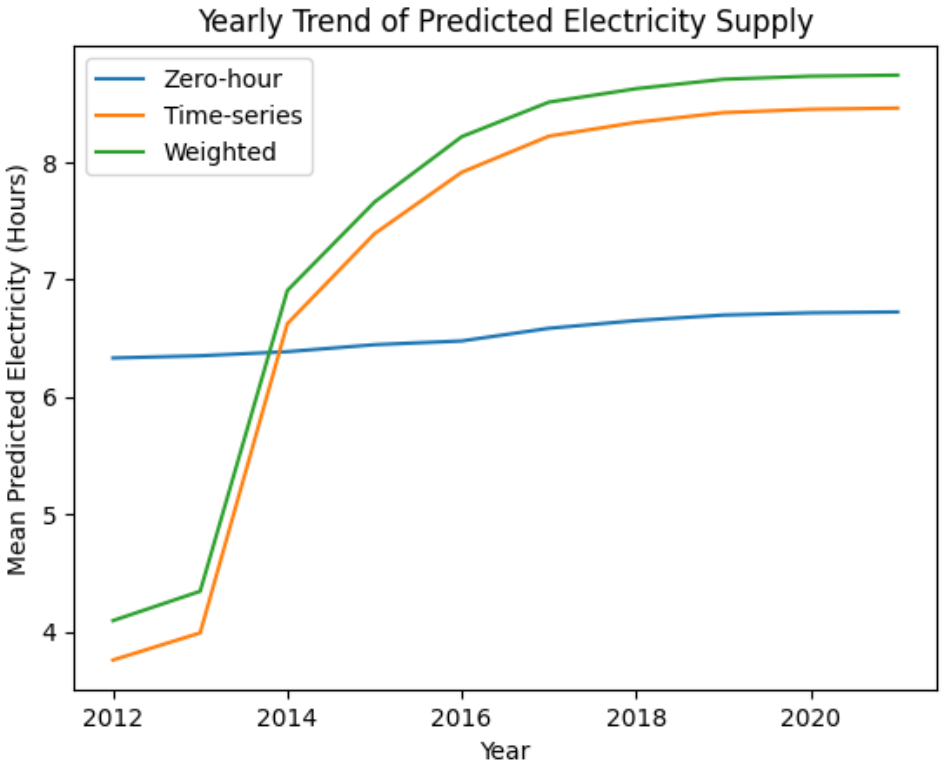


Figure C1: Yearly mean predicted electricity supply

The weighted model further refines this trend, producing smoother and slightly higher predictions over time. This suggests that incorporating both temporal structure and weighting allows the model to better capture the dynamics of electrification.

E.3.2 Distribution of Predictions

Figure C2 shows the distribution of predicted electricity supply across Bihar.

The zero-hour model exhibits a concentration of low predicted values, indicating over-reliance on constructed zero-hour observations. The time-series model shifts the distribution toward higher values, while the weighted model produces a more balanced and smoother distribution.

This indicates that the weighted model is better able to capture heterogeneity in electricity supply without being dominated by artificial low values.

E.3.3 Range of Predictions

Figure C3 illustrates the range of predicted electricity supply over time for the weighted model.

The weighted model maintains a reasonable spread between minimum and maximum values while avoiding extreme low predictions. The mean prediction increases steadily over time,

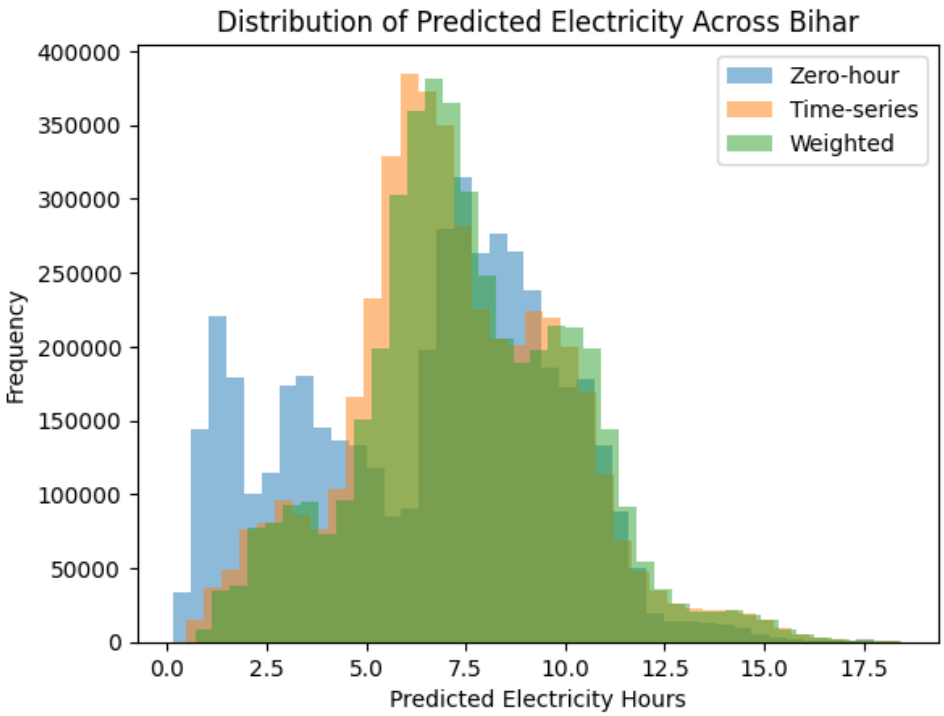


Figure C2: Distribution of predicted electricity supply across Bihar

indicating consistency with expected improvements in electricity access.

E.3.4 Summary Statistics

Model	Mean	Min	Max	Std
Zero-hour	6.54	0.17	17.77	3.25
Time-series	7.24	0.49	18.36	2.73
Weighted	7.54	0.70	18.47	2.71

Table C2: Summary statistics of predicted electricity supply

The results show a clear increase in mean predicted electricity supply from the zero-hour to the weighted model. The minimum predicted value also increases, indicating a reduction in unrealistic near-zero predictions. Additionally, the standard deviation decreases slightly, suggesting more stable and consistent predictions.

E.4 Discussion

The results highlight the importance of both data construction and modelling strategy. While the inclusion of zero-hour observations improves model fit, it introduces bias that can distort predictions. Time-series modelling improves temporal consistency, while the weighting strategy effectively reduces the influence of artificially constructed observations.

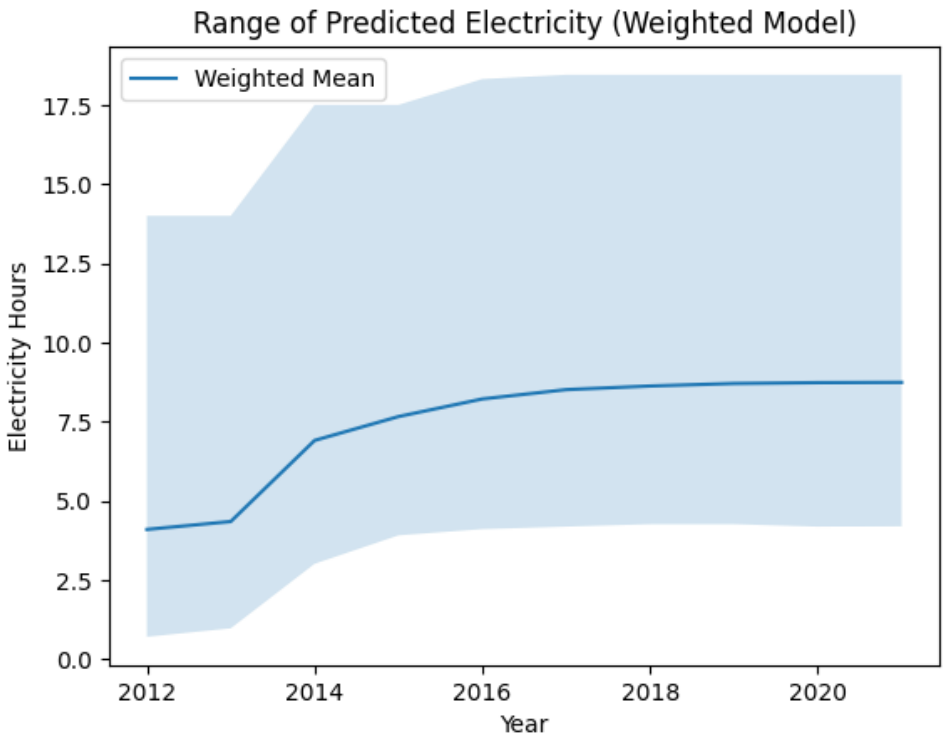


Figure C3: Range of predicted electricity supply (weighted model)

Overall, the weighted time-series model provides the most realistic and balanced predictions. Although it does not achieve the lowest prediction error, it produces more plausible distributions and temporal patterns of electricity supply, making it the preferred model for final estimation across Bihar.